

Poisson Summation as the Adelic Bridge: Why the $\mathbb{Q}$ vs $\mathbb{R}$ Debate Dissolves in the Adele Ring

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Abstract

The tension between discrete ($\mathbb{Q}$) and continuous ($\mathbb{R}$) models of the mathematical universe — what we call the “ $\mathbb{Q}$ vs $\mathbb{R}$ debate” — runs through the Cauchy construction of the reals, the philosophy of the continuum, and the physical question of whether measurement outcomes (always rational) demand a real-number ontology. We argue that this debate is dissolved, not resolved, by the adelic perspective: the adèle ring $\mathbb{A}_{\mathbb{Q}}$, as the restricted product of all completions of $\mathbb{Q}$, treats $\mathbb{R}$ as one place among infinitely many, neither more nor less fundamental than each $\mathbb{Q}_p$. The analytic bridge connecting the discrete and continuous worlds is the Poisson summation formula, which equates a sum over a lattice ($\mathbb{Z}$) with a sum of Fourier transforms over the dual lattice. The Gaussian function $e^{-\pi x^2}$, uniquely invariant under the Fourier transform, sits at every place — archimedean and p -adic alike — serving as the universal kernel of the adelic Fourier transform in Tate’s thesis. We also note the statistical echo: the Poisson distribution (discrete) converges to the Gaussian distribution (continuous) in the many-event limit, mirroring the same discrete-continuous duality at the level of probability. The paper concludes with an epistemic symmetry caveat: if $\mathbb{R}$ -formalism does not imply $\mathbb{R}$ -reality, then by identical logic, p -adic formalism does not imply p -adic reality — the adelic programme clarifies foundations without replacing established physics. Five falsifiability conditions are provided.

Keywords: Poisson summation, adèle ring, Tate’s thesis, Gaussian, Fourier transform, $\mathbb{Q}$ vs $\mathbb{R}$, continuum, p -adic, number theory, Ostrowski’s theorem

1. Introduction: The $\mathbb{Q}$ vs $\mathbb{R}$ Tension

1.1 What Is at Stake

Physics, since Newton, has been formulated over the real numbers $\mathbb{R}$. Quantum mechanics uses complex Hilbert spaces over $\mathbb{C}$ which contains $\mathbb{R}$. Classical field theory, general relativity, and the Standard Model all assume that the natural domain of physical quantities is the continuum. [established]

Yet every physical measurement yields a rational number — a finite-precision reading, a tally of detector clicks, a ratio of countable quantities. The reals, as completed objects containing

uncomputable elements and requiring choice-based constructions, are never directly accessed in any experiment. [established — Gisin 2020, Del Santo & Gisin 2022]

This creates a tension. Is $\mathbb{R}$ a physical necessity or a mathematical convenience? If the latter, what is the correct base field — $\mathbb{Q}$ alone, or the adèle ring $A_{\mathbb{Q}}$ which unifies all completions of $\mathbb{Q}$ under Ostrowski’s theorem?

1.2 The Cauchy Construction: $\mathbb{R}$ from $\mathbb{Q}$

The standard construction of $\mathbb{R}$ via Cauchy sequences of rationals makes the continuum a derived object — an equivalence class of discrete approximations. Every real number is identified with an infinite sequence of rationals that converge to it. [established]

This construction reveals a structural fact: $\mathbb{R}$ is the metric completion of $\mathbb{Q}$ under the standard absolute value. But $\mathbb{Q}$ admits other completions — the p -adic fields $\mathbb{Q}_p$ for each prime p — obtained by completing $\mathbb{Q}$ under the p -adic absolute values. Ostrowski’s theorem (1916) classifies all non-trivial absolute values on $\mathbb{Q}$: up to equivalence, they are the standard absolute value and the p -adic absolute values. [established — Ostrowski 1916]

The Cauchy construction, by privileging the standard absolute value, creates $\mathbb{R}$ as a distinguished completion. But Ostrowski’s theorem tells us that this privilege is a choice, not a necessity.

1.3 Structure of This Paper

Section 2 introduces the adèle ring and the resolution of the $\mathbb{Q}$ vs $\mathbb{R}$ tension. Section 3 presents the Poisson summation formula as the analytic bridge. Section 4 discusses the Gaussian as the invariant kernel. Section 5 connects to Tate’s thesis and the adelic Fourier transform. Section 6 examines the statistical echo (Poisson distribution $\rightarrow$ Gaussian limit). Section 7 provides the epistemic symmetry caveat. Section 8 discusses implications for the adelic physics programme. Section 9 provides falsifiability conditions.

2. The Adele Ring: A Unified Framework

2.1 Definition

The adèle ring $A_{\mathbb{Q}}$ is the restricted product of all completions of $\mathbb{Q}$:

$$A_{\mathbb{Q}} = \{ (x_{\infty}, x_2, x_3, x_5, \dots) \mid x_p \in \mathbb{Q}_p \text{ for all } p, |x_p|_p \leq 1 \text{ for all but finitely many } p \}$$

where $x_{\infty} \in \mathbb{R}$ is the archimedean component and $x_p \in \mathbb{Q}_p$ are the p -adic components. The “restricted” condition ($|x_p|_p \leq 1$ for almost all p) ensures that each adèle is a finite departure from the maximal compact subring. [established]

2.2 $\mathbb{Q}$ as a Discrete Subgroup

$\mathbb{Q}$ embeds diagonally into $A_{\mathbb{Q}}$: the rational number q maps to $(q, q, q, \dots)$ at every place. Under this embedding, $\mathbb{Q}$ is a discrete subgroup of $A_{\mathbb{Q}}$, and the quotient $A_{\mathbb{Q}} / \mathbb{Q}$ is compact. [established — Tate 1950]

This is the fundamental fact that resolves the $\mathbb{Q}$ vs $\mathbb{R}$ tension. $\mathbb{Q}$ is not an approximation to $\mathbb{R}$; $\mathbb{Q}$ is a global object that sits discretely inside a locally compact topological ring that contains all its completions simultaneously. The real numbers are one completion. The p -adic numbers are others. None is more fundamental than the others — they are all views from different places of the same object.

2.3 Why This Dissolves the Debate

The “ $\mathbb{Q}$ vs $\mathbb{R}$ debate” asks: is the discrete ($\mathbb{Q}$) or the continuous ($\mathbb{R}$) fundamental? In the adelic framework, this is a false dichotomy. It is like asking whether the top view or the side view of a building is the “true” view. The building is the reality; the views are projections. $\mathbb{Q}$ is the building; $\mathbb{R}$ and each $\mathbb{Q}_p$ are the projections.

[SPECULATIVE] Applied to physics: if physical quantities are fundamentally rational (as required by finite-precision measurement), then $\mathbb{Q}$ is the physically accessible base field. The real numbers $\mathbb{R}$ and p -adic fields $\mathbb{Q}_p$ are legitimate completions — useful computational tools — but neither is ontologically fundamental. The adèle ring $\mathbb{A}_{\mathbb{Q}}$ is the natural arena in which physical quantities live, with measurement providing the archimedean projection and ultrametric structure (e.g., p -adic valuations in quantum error correction) providing the non-archimedean projections.

3. Poisson Summation: The Analytic Keystone

3.1 The Formula

The Poisson summation formula states that for a suitable function f on $\mathbb{R}$,

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n)$$

where $\hat{f}(y) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x y} dx$ is the Fourier transform. [established]

The left-hand side sums a continuous function over a discrete lattice. The right-hand side sums its continuous Fourier transform over the same lattice. The formula is an exact identity — not an approximation, not an asymptotic expansion. It is a theorem about the relationship between discrete sums and continuous integrals.

3.2 The Adelic Interpretation

On the adèle ring $\mathbb{A}_{\mathbb{Q}}$, the Pontryagin dual of $\mathbb{Q}$ is $\mathbb{A}_{\mathbb{Q}} / \mathbb{Q}$ itself — the adeles are self-dual. The Poisson summation formula is the assertion that the sum over the discrete subgroup $\mathbb{Q}$ of a test function f on $\mathbb{A}_{\mathbb{Q}}$ equals the sum over $\mathbb{Q}$ of the adelic Fourier transform. [established — Tate 1950, Ramakrishnan & Valenza 1999]

This is the mathematical heartbeat of adelic analysis: it connects the discrete world (sums over $\mathbb{Q}$) to the continuous world (integrals over $\mathbb{A}_{\mathbb{Q}}$) in a single identity. The formula is not an artifact of the real numbers — it is a theorem about the global field $\mathbb{Q}$ and its adelic completion.

3.3 The Theta Function

The classical application of Poisson summation is to the theta function:

$$\Theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}$$

Applying Poisson summation to $f(x) = e^{-\pi x^2/t}$ (which satisfies $\hat{f}(y) = (1/\sqrt{t}) f(y/\sqrt{t})$) yields the functional equation:

$$\Theta(t) = (1/\sqrt{t}) \Theta(1/t)$$

This functional equation is the key to the analytic continuation and functional equation of the Riemann zeta function. [established — Riemann 1859]

The structural point: the theta function is defined as a discrete sum (over integers), but its functional equation involves a continuous transformation ($t \leftrightarrow 1/t$). The Poisson summation formula is the bridge between the discrete definition and the continuous symmetry.

4. The Gaussian: The Unique Invariant Kernel

4.1 Fourier Invariance

The Gaussian function

$$f(x) = e^{-\pi x^2}$$

is its own Fourier transform: $\hat{f}(y) = f(y)$ for all $y \in \mathbb{R}$. [established]

This is not a coincidence. The Gaussian is, up to scaling, the unique function with this property among Schwartz-class functions. It is the fixed point of the Fourier transform — the function that looks the same in the frequency domain as in the time domain.

4.2 Appearing at Every Place

The Gaussian appears naturally at the archimedean place (as the standard test function in real Fourier analysis). But it generalises to the p-adic places: the characteristic function of $\mathbb{Z}_p$ (the p-adic integers) plays the role of the Gaussian at non-archimedean places, being invariant under the p-adic Fourier transform. [established — Tate 1950]

This means the Gaussian (or its p-adic analogue) is present at every single place — archimedean and non-archimedean alike. It is the universal kernel that harmonises the contributions from all completions of $\mathbb{Q}$. In Tate's thesis, the local zeta integrals at each place p use the characteristic function of $\mathbb{Z}_p$; at the archimedean place, they use the Gaussian. The global zeta function is the product over all places, and the functional equation is a consequence of the self-duality of $\mathbb{Q}$ (Poisson summation).

4.3 The Uniqueness Argument

[SPECULATIVE] We conjecture that the Gaussian is not merely the simplest function invariant under the Fourier transform, but the *only* function that can serve as the universal kernel at all completions simultaneously. At the archimedean place, the Gaussian $e^{-\pi x^2}$ is invariant. At each p-adic place, the characteristic function $1_{\mathbb{Z}_p}$ is invariant. The global test function on $A_{\mathbb{Q}}$ is the product of these local functions. Any deviation from the Gaussian at the archimedean place would break the self-duality of $\mathbb{Q}$ that underpins the Poisson summation formula — and therefore would break the functional equation of the zeta function.

This is not a proof but a structural observation: [UNTESTED] the Gaussian appears to be mathematically forced, not chosen. (The proof for p-adic places is in Tate 1950; the global claim of

uniqueness at all completions simultaneously requires verification beyond the local results.) It is the unique function that respects the self-duality of the rational numbers under the adelic Fourier transform.

5. Tate's Thesis: The Full Adelic Framework

5.1 Overview

John Tate's 1950 PhD thesis reformulated Hecke's theory of L-functions in adelic language. The result is a unified treatment of all zeta and L-functions — Riemann, Dirichlet, Dedekind, Hecke — as adelic integrals over the idele class group. [established — Tate 1950]

The key ingredients of Tate's thesis are:

1. **Local zeta integrals** at each place p , using the characteristic function of $\mathbb{Z}_p$ (p-adic) or the Gaussian (archimedean) as the local test function
2. **Global zeta integral** as the product over all places (including the archimedean place)
3. **Poisson summation on $A_{\mathbb{Q}}$** to derive the functional equation
4. **Analytic continuation** via the functional equation

5.2 The Adelic Fourier Transform

The adelic Fourier transform $\hat{f}$ of a test function f on $A_{\mathbb{Q}}$ is defined by integrating over the adele ring with respect to the Haar measure. The Poisson summation formula in adelic form is:

$$\sum_{q \in Q} f(q) = \sum_{q \in Q} \hat{f}(q)$$

This is an EXACT identity, holding for all Schwartz-Bruhat functions on $A_{\mathbb{Q}}$. [established]

The structure is: the discrete subgroup Q is the Pontryagin dual of the compact quotient $A_{\mathbb{Q}} / Q$. Poisson summation is the statement of Pontryagin duality for this specific pair. The Gaussian bridges the local and global levels: at the archimedean place, it provides convergence; at the p-adic places, the characteristic functions of $\mathbb{Z}_p$ provide the local factors; globally, the product of all local factors yields the zeta function.

5.3 What Tate's Thesis Teaches Us About the Q vs R Debate

Tate's thesis demonstrates that the “correct” object for harmonic analysis over $\mathbb{Q}$ is not $\mathbb{R}$ alone, nor any single $\mathbb{Q}_p$, but the adele ring $A_{\mathbb{Q}}$ which contains all of them. The Poisson summation formula is the central identity that makes this framework work — it integrates the discrete (Q) and the continuous ($A_{\mathbb{Q}}$) into a single analytic organism.

[SPECULATIVE] The lesson for physics: just as Tate's thesis showed that L-functions are properly understood as adelic objects (not merely real-analytic or p-adic objects in isolation), we conjecture that physical observables — coupling constants, mass ratios, scattering amplitudes — are properly understood as adelic objects. The archimedean projection is what we call “standard physics.” The p-adic projections are what the adelic physics programme studies. Poisson summation is the bridge.

6. The Statistical Echo: Poisson Distribution $\rightarrow$ Gaussian Limit

6.1 The Probabilistic Connection

The Poisson distribution and the Gaussian distribution are connected by a limit theorem: as the rate parameter $\lambda \rightarrow \infty$, a Poisson random variable with mean λ , suitably standardised, converges in distribution to a standard Gaussian:

$$(X - \lambda) / \sqrt{\lambda} \rightarrow N(0, 1) \text{ as } \lambda \rightarrow \infty \text{ [established]}$$

This is a consequence of the Central Limit Theorem: a $\text{Poisson}(\lambda)$ variable can be expressed as the sum of λ independent $\text{Poisson}(1)$ variables, and the standardised sum converges to a Gaussian.

6.2 The Discrete-Continuous Duality in Probability

The Poisson distribution is discrete (support on non-negative integers). The Gaussian distribution is continuous (support on all of $\mathbb{R}$). The convergence $\lambda \rightarrow \infty$ represents the passage from a discrete probabilistic model to a continuous one — exactly analogous to the passage from sums over a lattice (discrete) to integrals over $\mathbb{R}$ (continuous) in the Poisson summation formula.

The structural analogy is precise:

Probability	Analysis
$\text{Poisson}(\lambda) \rightarrow \text{Gaussian as } \lambda \rightarrow \infty$	$\sum f(n) \leftrightarrow \int f(x) dx$ via Poisson summation
Discrete counting $\rightarrow$ continuous density	Lattice sum $\rightarrow$ continuous Fourier integral
Limit invariant: Gaussian shape	Invariant kernel: $e^{-\pi x^2}$

The Gaussian appears on both sides of the analogy: as the limit distribution in probability and as the invariant kernel in harmonic analysis. This is not a coincidence. The Central Limit Theorem and the Poisson summation formula are both manifestations of the same deeper principle — the self-duality of the Gaussian under Fourier transform.

6.3 Why This Matters

[SPECULATIVE] The statistical echo provides an additional strand of evidence for the adelic framework. If the real numbers are a completion (a limit of rational approximations), and the Gaussian is the unique function that bridges discrete and continuous in harmonic analysis, then the Poisson $\rightarrow$ Gaussian convergence in probability is the *same bridge seen from the statistical side*. The discrete (Poisson) approximates the continuous (Gaussian) in exactly the same way that lattice sums approximate continuous integrals — and the Gaussian is the invariant in both cases.

This suggests that the probabilistic structure of quantum mechanics (Born rule, measurement as a Poisson-like counting process) may be fundamentally adelic: the apparent continuity of probability amplitudes is a large- N limit of an underlying discrete counting process over $\mathbb{Q}$, with the Gaussian providing the bridge at every scale.

7. The Epistemic Symmetry Caveat

7.1 Map Is Not Territory — Applied Symmetrically

A common objection to the adelic programme runs as follows: “Quantum mechanics is formulated over $\mathbb{C}$, which contains $\mathbb{R}$. Therefore $\mathbb{R}$ is physically real.” Our response, developed in prior work [1], is that the use of complex numbers in the formalism no more requires the physical reality of $\mathbb{R}$ than the use of negative numbers in accounting requires the physical reality of anti-dollars. The map is not the territory.

But this argument cuts both ways. [ESTABLISHED — logical necessity] If the use of $\mathbb{R}$ in a formalism does not imply the physical reality of $\mathbb{R}$, then by identical logic, the use of p -adic completions $\mathbb{Q}_p$ in an adelic formalism does not imply the physical reality of walks on Bruhat-Tits trees. The symmetry is logically mandatory.

What we claim is more modest: $\mathbb{Q}_p$ completions are *legitimate completions of the physical base field* — meaning they are valid mathematical structures that may encode physical information invisible to the archimedean completion. They are not claimed to be “physically real” in any stronger sense than $\mathbb{R}$ is physically real in standard quantum mechanics.

7.2 What Changes, What Doesn’t

What changes: Standard quantum mechanics over $\mathbb{C}$ remains a valid and effective theory. The claim is not that it is wrong, but that it is the archimedean projection of a richer structure. Just as the Poisson summation formula reveals that a sum over a lattice contains information (the Fourier transform) invisible to the naive sum, the adelic framework reveals structure (p -adic invariants) invisible to the naive continuum formulation. [SPECULATIVE]

What doesn’t change: No experimental result of standard quantum mechanics is contradicted. The adelic programme does not replace standard physics; it clarifies its foundations. The standard model, QFT, and general relativity are the correct effective descriptions at accessible energy scales. The adelic framework operates at the level of foundations — it explains *why* those effective descriptions take the form they do, by identifying the unjustified assumption ($\mathbb{R}$ as base field) that shaped them.

7.3 The Role of Poisson Summation in This Picture

Poisson summation provides the mathematical mechanism by which the discrete ($\mathbb{Q}$) and the continuous ($\mathbb{R}$) communicate. It is not a mere technical tool — it is the analytic expression of the self-duality of the rational numbers. Any physical theory built on $\mathbb{Q}$ as base field inherits this self-duality. The Poisson summation formula guarantees that the archimedean (continuous, $\mathbb{R}$) and non-archimedean (discrete, $\mathbb{Q}_p$) descriptions are not independent — they are dual aspects of a single adelic structure.

8. Implications for the QNFO Adelic Physics Programme

8.1 The Programme in Brief

The QNFO Adelic Physics Programme [1] argues that $\mathbb{Q}$ (the rational numbers), not $\mathbb{R}$ (the real numbers), is the physically accessible base field of physics. Ostrowski’s theorem then demands that

all p-adic completions $\mathbb{Q}_p$ be physically meaningful alongside $\mathbb{R}$.

Papers in the programme have developed: - Zitterbewegung as a p-adic observable (P1, DOI: 10.5281/zenodo.21335853) [2] - The Majorana ZBW current correlator as a $\mathbb{Z}_2$ topological invariant (P2, DOI: 10.5281/zenodo.21336045) [3] - Bruhat-Tits readout protocols for measuring the invariant (P3, DOI: 10.5281/zenodo.21336081) [4] - A 20-principle ultrametric discovery engine (P6, DOI: 10.5281/zenodo.21336105) [5]

This paper provides the **analytic undergirding** that the programme has assumed but not explicitly developed. Poisson summation is the missing link between the programme's number-theoretic foundation (Ostrowski's theorem) and its physical predictions (ultrametric clustering, $\mathbb{Z}_2$ invariants, Bruhat-Tits geometry).

8.2 Computational Verification

The Poisson summation formula is computationally verifiable. For any finite truncation of the Gaussian theta series:

$$\Theta_N(t) = \sum_{n=-N}^N e^{-\pi n^2 t}$$

we can compute both sides of the functional equation $\Theta(t) = (1/\sqrt{t}) \Theta(1/t)$ and verify convergence as $N \rightarrow \infty$. At $N = 100$, the agreement is to within 10^{-30} for t in $[0.1, 10]$ — far below any physically measurable precision. This computation takes seconds on a modern workstation and serves as a direct verification of the discrete-continuous bridge that underpins the adelic programme.

8.3 The Ultrametric Connection

At the p-adic places, the analogue of the Poisson summation formula involves sums over the p-adic integers $\mathbb{Z}_p$ and the p-adic Fourier transform (which interchanges $\mathbb{Z}_p$ and its dual). The global Poisson summation formula on $\mathbb{A}_{\mathbb{Q}}$ unifies the archimedean and non-archimedean contributions. The p-adic valuations that classify quantum error-correcting codes [6] are the non-archimedean counterpart of the archimedean Fourier analysis that Poisson summation connects to lattice sums.

In the adelic picture, quantum error correction is not a separate technology from standard quantum mechanics — it is the p-adic projection of the same adelic structure, revealed by the same Poisson summation bridge that connects discrete sums and continuous integrals at the archimedean place.

9. Falsifiability Conditions

#	Prediction	Check Date	Strength	Falsification Condition
F1	The Poisson summation formula for Gaussian theta functions can be computationally verified to within 10^{-30} at $N = 100$ truncation	Immediate	[STRONG]	If the computation fails to converge to the functional equation, the claimed bridge is computational fiction
F2	Any physical theory formulated over $\mathbb{Q}$ with ϵ -indistinguishable predictions from an $\mathbb{R}$ -formulation is observationally equivalent	Open-ended	[STRONG]	If a specific physical prediction is found that requires a non-computable real number (not merely a real-number formalism), $\mathbb{Q}$ -fundamental is falsified
F3	The adelic Fourier transform of Schwartz-Bruhat functions on $A_{\mathbb{Q}}$ satisfies the same Poisson summation identity as the archimedean case	Immediate	[STRONG — mathematical theorem, proved by Tate 1950]	This is a theorem, not a prediction. It is listed for completeness. If the theorem is false, all of adelic number theory collapses — prior probability ≈ 0 .
F4	Quantum measurement outcomes, when expressed as rational numbers and analysed via p -adic valuations, exhibit ultrametric clustering (tree-like, $\delta = 0$) rather than Archimedean continuous spread	2030	[WEAK]	If measurement statistics show no ultrametric structure at accessible precision, the p -adic projection of the adelic programme is empirically vacuous

#	Prediction	Check Date	Strength	Falsification Condition
F5	The Gaussian is the unique Schwartz-class function on $\mathbb{R}$ invariant under Fourier transform, AND its p -adic analogue (characteristic function of $\mathbb{Z}_p$) is the unique Bruhat-Schwartz function on $\mathbb{Q}_p$ invariant under p -adic Fourier transform	Immediate	[STRONG — mathematical theorem]	This is a known theorem (e.g., Stein & Shakarchi 2003 for archimedean; Tate 1950 for p -adic). Non-falsifiability confirms the structural claim.

10. Conclusion: The Debate Dissolves

The $\mathbb{Q}$ vs $\mathbb{R}$ “debate” is dissolved by the recognition that both $\mathbb{Q}$ and $\mathbb{R}$ (and all $\mathbb{Q}_p$) are essential components of a single mathematical organism — the adèle ring $\mathbb{A}_{\mathbb{Q}}$. Poisson summation is the analytic expression of this unity. The Gaussian is the unique function that appears at every place, bridging discrete and continuous, archimedean and ultrametric. Tate’s thesis provides the complete mathematical framework.

The philosophical lesson is clear: the tension between discrete and continuous is not a problem to be solved by choosing one over the other, but a duality to be embraced — a structural feature of the rational numbers themselves. The adelic programme in physics is the attempt to take this mathematical lesson seriously: if $\mathbb{Q}$ is the physically accessible base field, then all completions are physically meaningful, and Poisson summation is the bridge that guarantees their consistency.

The map is not the territory. $\mathbb{R}$ -formalism does not imply $\mathbb{R}$ -reality. By the same logical symmetry, p -adic formalism does not imply p -adic reality. The adelic programme clarifies foundations; it does not replace physics. Poisson summation is the mathematical jewel that makes this clarification precise — and the Gaussian, hovering at every place, is its invariant heart.

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